

Utkarsh Kumar

Software Engineer | Machine Learning | Systems Design | Computer Vision
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Summary

Third-year Computer Science student and published IGARSS 2026 researcher with production experience building ML systems at scale, a 110M+ resume search engine, GNN-based fraud detection, and real-time computer vision pipelines. Two-time national hackathon winner with a strong foundation in algorithms, distributed systems, cloud deployment, and applied ML research.

Education

Manipal Institute of Technology, Bengaluru B.Tech in Computer Science (Data Science)	GPA: 9.3/10 2024 – 2028
Indian Institute of Technology, Guwahati B.Sc Honours in Artificial Intelligence and Data Science	GPA: 8.3/10 2024 – 2028

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL

AI / ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), PyTorch, Scikit-Learn, XGBoost, LightGBM, Ensemble Learning, GraphSAGE, PyTorch Geometric, Hugging Face, LangChain, OpenAI API, FAISS, BM25, Computer Vision, OpenCV, DINOv2, spaCy, NetworkX

Backend / Systems: FastAPI, Flask, Node.js, Next.js, React, PostgreSQL, Redis, MongoDB, SQLAlchemy, Pydantic, Docker, Kubernetes, Microservices, REST APIs, WebSockets, GraphQL

Cloud / DevOps: Azure (AKS, AI Search), AWS (EC2, S3), GCP, Supabase, Vercel, Git, CI/CD, Linux

Data / Tools: Pandas, NumPy, D3.js, Microsoft Presidio, Prometheus, Grafana, Spatial Analytics, Chart.js

Experience

AI / Software Engineering Intern, Provaantech Apr 2026 – August 2026
GitHub | Demo

- Engineered an AI Candidate Discovery Engine scanning 110M+ resumes, returning top 10 ranked matches in under 10 seconds.
- Built hybrid retrieval pipeline combining OpenAI embeddings, Azure AI Search, BM25 ranking, and semantic re-ranking.
- Integrated GPT-4o-mini to auto-generate candidate fit explanations for each shortlisted match.

Software Engineering Intern, AVSS Tech Dec 2025 – Jan 2026

- Built a real-time face recognition pipeline achieving 92%+ accuracy for enterprise access control.
- Reduced inference latency by 35% via adaptive frame batching, frame skipping, and hardware-level optimizations.
- Designed modular detection-and-verification pipeline with graceful fallback logic for edge-case inputs.

Researcher, IEEE GRSS, Manipal Institute of Technology Jan 2025

- Built a two-stage hurdle stacking ensemble across 210 weather stations using 15 years of NASA POWER reanalysis data.
- Achieved $R^2=0.581$, MAE=2.17 mm, Skill Score=0.510; enforced leakage-free chronological CV with OOF meta-learning.

Projects

PII Redaction Microservice – Privacy Engineering Demo

- Built a zero per-request-cost PII redaction service for resume PDFs using Microsoft Presidio, spaCy NER, and custom regex recognizers for Indian-specific formats (PAN, Aadhaar, +91 numbers).
- Deployed on Azure AKS via FastAPI, Docker, and Kubernetes; handles high-throughput batch redaction with PyMuPDF and Redis caching.
- Instrumented with Prometheus and Grafana for real-time monitoring, on a typed, zero-disk-I/O architecture using Pydantic v2 for validation.

CrowdWisdom.AI – Multi-Agent LLM System GitHub | Live Demo

- Built a multi-agent orchestration system for automated workflow execution, improving scheduling efficiency by 40%.
- Integrated ElevenLabs Voice AI for real-time conversational interaction with dynamic LLM-driven agent routing.

Awards & Achievements

AgniSense – Predictive Fire Risk & Operational Command Platform GitHub

- Top Winner, National-Level Hackathon (Rs. 75,000 Prize).**
- Integrated 5 municipal datasets into a unified feature store; FastAPI + Uvicorn ML backend processes 1.7+ GB telemetry for real-time Khatra Score inference.

Parakh.AI – Edge-Native Industrial Quality Control

- Top 3, National-Level Hackathon (Rs. 20,000 Prize).**
- Built a zero-shot industrial anomaly detection pipeline using WideResNet-50, DINOv2, and FAISS similarity search, achieving 99.1% AUROC.
- Deployed on-device for real-time industrial QC with no cloud dependency; used Mistral-7B to auto-generate defect explanations.

SmurfPakad – Blockchain Fraud Detection via GNNs GitHub | Demo

- Top 3, State-Level Hackathon (Rs. 20,000 Prize).**
- Built a GraphSAGE-based GNN to identify smurfing and money laundering patterns across blockchain transaction graphs.
- Delivered full-stack solution with FastAPI, PostgreSQL, Docker, and D3.js force-directed graph visualizations.